Supplementary Material for "Cardinality Coverage in Evolutionary Feature Selection: A Budget-Aware Study of Initialization and Search Operators"

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The tables and the figure below give the effect sizes, per-dataset results, and the classifier comparison that the paper summarizes. They are generated by `make_tables.py` from the CSV files in `evidence/H13-E01/tables/` and `evidence/R1/tables/` of the paper's repository (`https://github.com/klemenkenda/fastener-p2-cardinality-coverage`). The files in `evidence/R1/` are computed by `review_supplement.py` from the distilled run logs of `https://github.com/klemenkenda/fastener-experiments`. Group names, configurations, and measures are those of the paper; OBL is the opposition-based start and MIIP our implementation of NSGAII-MIIP.

Table S1: Principal paired comparisons in the benchmark, datasets as units: mean difference, 95% bootstrap interval over datasets, matched-pairs rank-biserial correlation r, and Holm-corrected Wilcoxon p over the three pairs of configurations within a group and measure. This family differs from that of Table 3 of the paper (four groups), so the reach comparison of the two starts has slightly different corrected values there. Time to reach: restricted mean share of the budget spent before the archive holds a subset with at most ten features. Largest subset: number of features of the largest subset on the returned front

	Decision tree				Random forest				LightGBM			
Comparison	Mean	95% CI	r	p	Mean	95% CI	r	p	Mean	95% CI	r	p
Reach, sparse − OBL	+0.680	[+0.507, +0.840]	+1.00	<0.001	+0.635	[+0.471, +0.782]	+1.00	<0.001	+0.543	[+0.364, +0.721]	+1.00	<0.001
Reach, MIIP − OBL	+0.043	[+0.007, +0.090]	+1.00	0.125	+0.075	[+0.019, +0.150]	+1.00	0.016	+0.081	[+0.015, +0.170]	+1.00	0.062
Reach, sparse − MIIP	+0.641	[+0.469, +0.800]	+1.00	<0.001	+0.537	[+0.369, +0.700]	+1.00	<0.001	+0.464	[+0.286, +0.643]	+1.00	<0.001
AUF_{10}, sparse − OBL	+0.491	[+0.374, +0.601]	+0.94	<0.001	+0.477	[+0.361, +0.586]	+0.93	<0.001	+0.415	[+0.290, +0.539]	+0.85	<0.001
AUF_{10}, MIIP − OBL	+0.058	[+0.015, +0.112]	+0.87	0.005	+0.057	[+0.007, +0.111]	+0.59	0.031	+0.083	[+0.025, +0.150]	+0.62	0.029
AUF_{10}, sparse − MIIP	+0.437	[+0.315, +0.556]	+0.89	<0.001	+0.393	[+0.273, +0.511]	+0.95	<0.001	+0.332	[+0.202, +0.465]	+0.64	0.005
AUF_{20}, sparse − OBL	+0.482	[+0.361, +0.598]	+0.94	<0.001	+0.465	[+0.344, +0.579]	+0.92	<0.001	+0.402	[+0.273, +0.533]	+0.83	<0.001
AUF_{d}, sparse − OBL	+0.140	[+0.096, +0.183]	+0.89	<0.001	+0.127	[+0.080, +0.174]	+0.82	<0.001	+0.085	[+0.046, +0.126]	+0.66	0.006
Best F1, sparse − OBL	−0.007	[−0.018, +0.007]	−0.58	0.013	−0.029	[−0.043, −0.016]	−0.76	<0.001	−0.022	[−0.032, −0.013]	−0.87	<0.001
Best F1, sparse − MIIP	−0.012	[−0.028, +0.002]	−0.51	0.036	−0.021	[−0.037, −0.006]	−0.61	0.004	−0.021	[−0.036, −0.007]	−0.67	0.003
Largest subset, sparse − OBL	−348.7	[−591.3, −157.5]	−0.99	<0.001	−2632.6	[−6927.3, −225.5]	−0.99	<0.001	−120.6	[−189.0, −61.3]	−0.98	<0.001
Time to reach, sparse − OBL	−0.771	[−0.888, −0.634]	−1.00	<0.001	−0.778	[−0.881, −0.658]	−1.00	<0.001	−0.724	[−0.847, −0.587]	−1.00	<0.001
Time to reach, MIIP − OBL	−0.044	[−0.081, −0.015]	−1.00	0.002	−0.065	[−0.110, −0.030]	−1.00	<0.001	−0.070	[−0.114, −0.033]	−1.00	<0.001

Table S2: Split of the AUF difference (sparse minus OBL start, $K = 10$) into reach and quality parts (Eq. 2 of the paper), by group. Mean over datasets of Δ, ρ, and κ; signed share S and absolute share A (Eq. 3), in percent with 95% bootstrap intervals over datasets; the share of the net difference that comes from limits the OBL start never reaches on a dataset; and the numbers of datasets with a negative and a positive quality part

Group	Data	$\bar{\Delta}$	$\bar{\rho}$	$\bar{\kappa}$	S	A	OBL never	$\kappa<0/\kappa>0$
Decision tree (designed)	8	+0.623	+0.624	−0.001	100.1 [99.9, 100.5]	99.8 [99.4, 100.0]	99.2 [97.0, 100.5]	1/1
Decision tree	30	+0.491	+0.491	+0.000	100.0 [99.8, 100.2]	99.7 [99.4, 99.9]	93.8 [85.8, 99.4]	5/5
Random forest	34	+0.477	+0.483	−0.006	101.2 [99.5, 104.3]	98.1 [95.2, 99.7]	93.1 [83.8, 100.2]	6/7
LightGBM	28	+0.415	+0.415	+0.000	100.0 [99.3, 100.8]	99.2 [98.3, 99.8]	88.4 [74.6, 97.4]	6/7

Table S3: Matched-size difference in test F1, sparse minus OBL start, for every dataset on which it is defined. $|M|$: number of limits up to ten that both starts reach in at least three of five seeds; m: matched-size difference; seed-paired: number of seeds with a shared reached limit, mean, minimum, and maximum of the per-seed differences. Datasets with at most 50 test rows are left out only in the exploratory analysis

						Seed-paired			
Group	Dataset	d	Test rows	$\|M\|$	m	Seeds	Mean	Min	Max
Decision tree (designed)	satimage	36	1286	10	−0.007	5	−0.007	−0.009	−0.003
Decision tree	PenDigits	16	2199	10	−0.000	5	−0.000	−0.002	+0.002
Decision tree	ItalyPowerDemand	24	219	10	+0.004	5	+0.004	+0.000	+0.018
Decision tree	Crop	46	4800	10	+0.004	5	+0.004	−0.003	+0.009
Decision tree	ProximalPhalanxOutlineCorrect	80	178	10	−0.009	5	−0.009	−0.024	+0.014
Decision tree	TwoLeadECG	82	232	2	−0.009	5	−0.007	−0.047	+0.035
Decision tree	Libras	90	75	10	−0.001	5	−0.001	−0.050	+0.055
Decision tree	ElectricDevices	96	3327	8	+0.005	5	+0.008	−0.009	+0.022
Decision tree	MedicalImages	99	229	10	−0.009	5	−0.009	−0.025	+0.016
Decision tree	ECG5000	140	1000	6	+0.002	4	+0.003	−0.008	+0.010
Decision tree	Adiac	176	156	9	+0.004	4	−0.008	−0.055	+0.011
Random forest	PenDigits	16	2199	10	−0.001	5	−0.001	−0.002	+0.000
Random forest	ItalyPowerDemand	24	219	10	−0.001	5	−0.001	−0.007	+0.009
Random forest	Crop	46	4800	10	+0.000	5	+0.000	−0.003	+0.004
Random forest	ProximalPhalanxOutlineCorrect	80	178	10	+0.001	5	+0.001	−0.015	+0.023
Random forest	TwoLeadECG	82	232	6	+0.022	5	+0.021	+0.011	+0.027
Random forest	Libras	90	75	10	+0.011	5	+0.011	−0.013	+0.068
Random forest	ElectricDevices	96	3327	3	+0.008	5	+0.008	−0.004	+0.022
Random forest	MedicalImages	99	229	10	+0.007	5	+0.007	−0.006	+0.016
Random forest	ECG5000	140	1000	6	+0.001	4	+0.001	−0.004	+0.005
Random forest	Adiac	176	156	8	+0.028	4	+0.021	−0.017	+0.054
Random forest	Strawberry	235	196	9	−0.009	4	−0.015	−0.024	−0.005
Random forest	Coffee	286	11	10	−0.138	5	−0.138	−0.258	+0.053
Random forest	BeetleFly	512	8	10	−0.106	5	−0.106	−0.494	+0.254
LightGBM	PenDigits	16	2199	10	+0.000	5	+0.000	−0.000	+0.000
LightGBM	ItalyPowerDemand	24	219	10	+0.005	5	+0.005	−0.006	+0.020
LightGBM	Crop	46	4800	10	−0.000	5	−0.000	−0.004	+0.004
LightGBM	ProximalPhalanxOutlineCorrect	80	178	10	−0.002	5	−0.002	−0.013	+0.011
LightGBM	TwoLeadECG	82	232	5	+0.004	5	+0.003	−0.022	+0.020
LightGBM	Libras	90	75	10	−0.007	5	−0.007	−0.042	+0.022
LightGBM	ElectricDevices	96	3327	4	+0.007	5	+0.008	+0.000	+0.016
LightGBM	MedicalImages	99	229	10	−0.001	5	−0.001	−0.011	+0.007
LightGBM	ECG5000	140	1000	6	+0.001	4	+0.000	−0.005	+0.010
LightGBM	Adiac	176	156	4	+0.036	4	+0.025	−0.042	+0.081
LightGBM	Strawberry	235	196	10	−0.005	5	−0.005	−0.016	+0.020
LightGBM	Coffee	286	11	10	+0.000	5	+0.000	−0.091	+0.091
LightGBM	BeetleFly	512	8	10	−0.029	5	−0.029	−0.267	+0.140

Table S4: Finished runs out of planned runs, by group and configuration

Group	Configuration	Finished	Planned	Share
Decision tree (designed)	OBL start	40	40	100%
Decision tree (designed)	Sparse start	40	40	100%
Decision tree	OBL start	164	195	84%
Decision tree	Sparse start	165	195	85%
Decision tree	NSGAII-MIIP	149	195	76%
Random forest	OBL start	170	195	87%
Random forest	Sparse start	195	195	100%
Random forest	NSGAII-MIIP	160	195	82%
LightGBM	OBL start	134	195	69%
LightGBM	Sparse start	194	195	99%
LightGBM	NSGAII-MIIP	133	195	68%

Table S5: Number of features at which half of the runs reach ten features, from the logistic model of reach on $\log_2 d$ and $\log_2$ of the budget, at the median budget of 6,000 fits; 95% percentile interval over 2,000 bootstrap resamples of datasets (resamples in which all runs agree are skipped). Descriptive locations for these datasets and budgets

Group	Configuration	d_{50}	95% CI	Resamples
Decision tree (designed)	OBL start	172	[92, 174]	1826
Decision tree	OBL start	163	[122, 202]	2000
Decision tree	NSGAII-MIIP	208	[159, 253]	2000
Random forest	OBL start	212	[145, 277]	2000
Random forest	NSGAII-MIIP	268	[224, 343]	2000
LightGBM	OBL start	226	[148, 300]	2000
LightGBM	NSGAII-MIIP	266	[210, 302]	2000

Table S6: Pilot: fits until the archive of an OBL run with ten times the budget first holds a subset with at most ten features; median and range over five seeds, in multiples of the benchmark budget, and per feature

Dataset	d	Budget	Median fits	Range	Budgets	Fits per feature
Adiac	176	12,000	6,534	6,128–8,452	0.54	37
Strawberry	235	6,000	9,292	5,472–16,662	1.55	40
Earthquakes	512	12,000	47,830	46,324–60,611	3.99	93
Computers	720	12,000	85,239	74,518–87,064	7.10	118

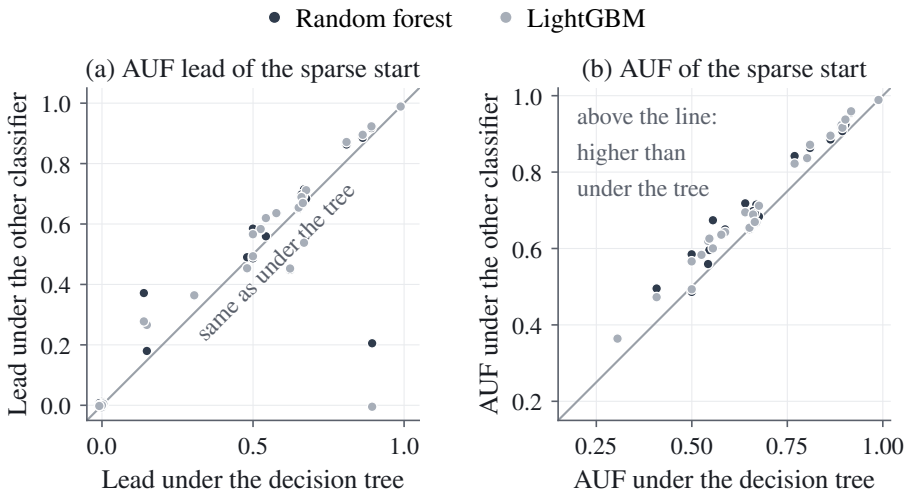


Figure S1: The classifier changes the scores but hardly the comparison (Appendix A of the paper). Each point is one of 26 datasets, under the random forest or LightGBM against the decision tree. (a) The AUF lead of the sparse start over the opposition-based start stays close to the diagonal. (b) The sparse start's own AUF lies above it: both other classifiers score higher than the tree.